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Auger recombination in 4H-SiC: Unusual temperature behavior

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The band-to-band Auger recombination in 4H-SiC material is studied using a time-resolved photoinduced absorption technique. The Auger recombination coefficient is derived from the kinetics of electron-hole plasma in heavily doped *n*-type 4H-SiC and in low-doped 4H-SiC epitaxial layers in the temperature interval 300–565 K. Within this range, its value decreases from $\gamma_3 = (7 \pm 1) \times 10^{-31} \text{ cm}^6 \text{ s}^{-1}$ to $\gamma_3 = (4 \pm 1) \times 10^{-32} \text{ cm}^6 \text{ s}^{-1}$. The observed pronounced reduction of Auger recombination rate with temperature is correlated to temperature dependent threshold energy of Auger process. © 1997 American Institute of Physics. [S0003-6951(97)04048-5]

Excess carrier lifetime is one of the key parameters governing semiconductor device performance. Band-to-band recombination, represented by radiative and Auger processes, are inherent even to a perfect crystal and provide the ultimate limit to carrier lifetimes. The indirect band-gap 4H-SiC polytype is a favorable semiconductor material for usage in high-power, high-temperature, high-frequency, and high-radiation environments.¹ In spite of successful device applications, numerous physical parameters that govern the optical and electronic behavior, as well as the high speed response of devices, are still unknown. Among them the Auger recombination (AR) rate is of particular significance as it determines the carrier lifetime in heavily doped emitters and sets an upper limit of the reachable injection level during the current conduction in bipolar devices. Thus, a high AR rate could effectively limit the use of 4H-SiC material for certain high-power device applications, and its correct value needs to be determined urgently. To the authors knowledge, there are neither experimental, nor theoretical evaluation of the band-to-band AR rates in SiC so far. We report here the first experimental measurements of Auger recombination coefficients for 4H-SiC material in a wide temperature range.

We have investigated 30 μm thickness *n*-type 4H-SiC epitaxial films, doped to $\sim 5 \times 10^{15} \text{ cm}^{-3}$ during the chemical vapor deposition on heavily nitrogen doped ($\geq 10^{19} \text{ cm}^{-3}$) substrates.² A series of substrate wafers with the nominal dopant concentration ranging from 3 to $8 \times 10^{18} \text{ cm}^{-3}$ were also examined. In addition, an auxiliary 300- μm -wide specimen has been sawed from the epilayer-substrate structure, and the side-walls of the strip were polished to optical quality.

The applied pump and probe technique is based on time-resolved measurements of absorption changes $\Delta\alpha$, induced by the excess density N of electron-hole (*e-h*) pairs. The plasma dynamics is experimentally deduced from $N(t) = \ln[I_0/I(t)](d\sigma_{eh})^{-1}$, where d is the probed path, I_0 is the

incoming, $I(t)$ is the transmitted probe beam intensities, and σ_{eh} stands for the free carrier absorption cross section. We have used 355 nm wavelength and 2.5 ns duration pulses from a Nd:YAG laser (Coherent, Infinity) as photoexcitation source. Absorption changes were probed by 1300 nm wavelength cw laser beam, focused to a diameter of $\sim 5 \mu\text{m}$ on the surface. The transients were detected by 0.5 ns rise time photoreceiver and processed by the digital 1 GHz bandwidth oscilloscope. In order to suppress interference effects and minimize reflection losses, the probe beam is *p*-polarized and its inclination to the surface is close to the Brewster angle. The probing could be provided in both perpendicular and parallel directions with respect to the photoexcitation beam. The sample holder is mounted on a computer-guided micropositioner with 1 μm translation increment and renders stabilized sample heating in the 300–565 K temperature range.

The AR rate can be determined from carrier concentration transients measured in two different ways. The first method involves optical excitation of a highly dense *e-h* plasma in a low-doped semiconductor, and refers to measurements of the excess carrier lifetime in a high-injection regime. The alternative way is to evaluate the minority carrier lifetimes in a heavily doped semiconductor under low-injection conditions, i.e., keeping the excess carrier density well below the doping level. In our study the initial excess carrier density was estimated on the basis of the measured photon flux and absorption of the pump radiation α_{ex} , assuming a unity quantum efficiency for the generation of *e-h* pairs by each absorbed photon. Since the major possible error source is α_{ex} , a careful examination of the absorption at pump wavelength and a calibration of σ_{eh} as a function of carrier density and temperature have preceded the main sequence of experiments. The actual α_{ex} was estimated applying a strip-shaped sample in perpendicular optical geometry—photoexcitation was provided in normal direction to the surface, while a focused probe beam was aligned along the epilayer. The pump penetration profiles were derived from the measured variation of the peak carrier density with depth, as illustrated in Fig. 1. By fitting the slopes recorded

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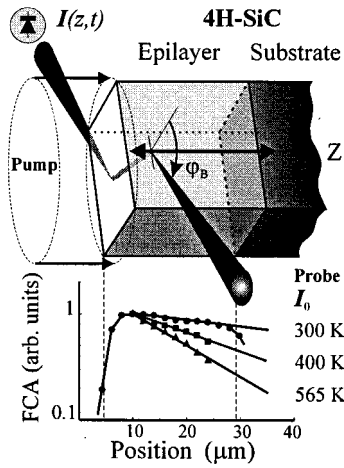


FIG. 1. Experimental configuration for depth-resolved photoinduced absorption measurements and normalized distributions of photoexcited carriers in 4H-SiC epilayer at different temperatures (bottom).

at different temperatures, the $\alpha_{\text{ex}}(T)$ dependence was then ascertained. We have observed a gradual increase of absorption for $\lambda_{\text{ex}}=355$ nm wavelength in the temperature interval 300–565 K, which for practical needs can be approximated by $\alpha_{\text{ex}}(T)=165+5\times[T-300]^{0.76}$.

In the following high-injection experiments the carrier recombination parameters were extracted by fitting the absorption transients with those simulated numerically. To separate and quantify numerous fitting parameters dependent upon material properties and injection level, we have exercised measurements of carrier dynamics (i) in parallel geometry and (ii) at different pump fluences. The first issue essentially simplifies simulation of photocarrier relaxation, since one-dimensional approach to carrier transport can be applied. In fact, once a focused probe beam is superimposed on the surface with uniformly flat ~ 6 mm pump spot, the lateral carrier diffusion can be discarded. At this point, the temporal and spatial evolution of photocarriers is described by rate equation:

$$\frac{\partial N(z,t)}{\partial t} = D \frac{\partial^2 N}{\partial z^2} - \gamma_1 N - \gamma_2 N^2 - \gamma_3 N^3 + G(z,t),$$

with the standard boundary conditions at the $z=0, l$ surfaces given by $\partial N(z,t)/\partial z = [S_{0,1}/D]N(z,t)$. Here $N(z,t)$ is the excess carrier density at a distance z from the surface and time moment t , l is thickness of the layer, D is the ambipolar diffusion coefficient, G is carrier generation rate, S_0 and S_1 are surface and interface recombination velocities. Coefficients γ_1 , γ_2 , and γ_3 specify monomolecular, bimolecular, and Auger recombination rates, respectively. Second, the accuracy of parameter extraction can be enhanced by simultaneous analysis of absorption transients obtained at different excitation intensities. In this case additional constraints in variation of fit parameters arise due to the fact that lowly excited carrier relaxation is determined mainly by density independent S , D , and γ_1 parameters. Once their values are prefixed, the fitting routine is adapted for the high-excitation decays to obtain the complete set of recombination param-

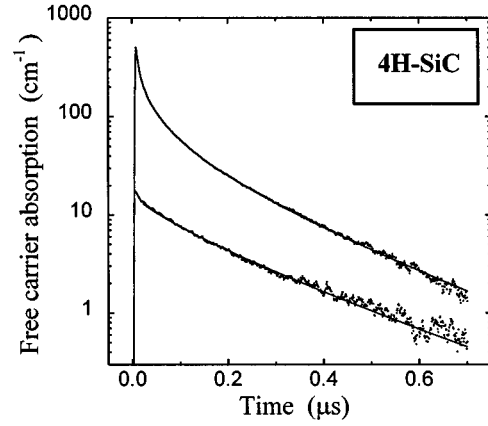


FIG. 2. Photoinduced absorption transients for $I=130$ and 4 mJ/cm² excitation intensities, probed at $\lambda_p=1.3$ μm wavelength. Points are experimental data, solid curves are numerically simulated fits.

eters. This procedure is illustrated in Fig. 2, where typical absorption transients and their numerical fits at different pump intensities are presented. The full-scale variation of instantaneous lifetime versus excess carrier density is shown in Fig. 3. The experimental data are presented by dots; the solid curve is simulated numerically, applying the following parameters: $\gamma_1=3.85\times 10^6$ s⁻¹, $\gamma_2=1.5\times 10^{-12}$ cm³ s⁻¹, $\gamma_3=7\times 10^{-31}$ cm⁶ s⁻¹, $S_0=4\times 10^4$ cm s⁻¹, $S_1=1\times 10^4$ cm s⁻¹, and $D=3$ cm² s⁻¹. Dotted lines in the plot indicate the quantified recombination rates. At low carrier concentrations γ_1 settles the asymptotic lifetime, apparently limited by Shockley–Reed–Hall defect centers in combination with processes of carrier diffusion and recombination on the film boundaries. It should be noted that a sizable bimolecular recombination rate γ_2 needs to be invoked to fit the experimental data, which is presumably contributed by both band-to-band radiative and trap-assisted AR mechanisms. Since electron and hole densities in a highly-excited plasma are the same, both the e - e - h and h - h - e transitions occur simultaneously, resulting in a combined AR rate $\gamma_3=\gamma_3^{eh}$

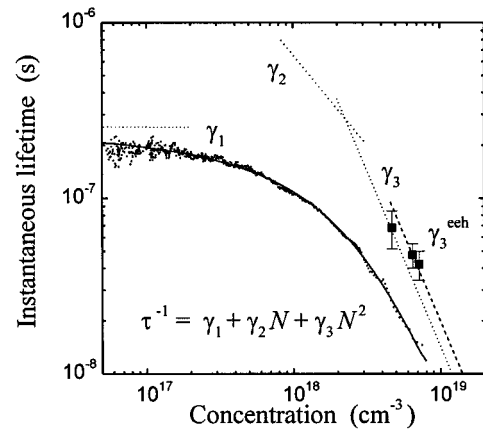


FIG. 3. Instantaneous lifetime as a function of carrier density. Dots are experimental data, solid curve is a numerical fit. Dotted lines conform to the extracted recombination rates: $\gamma_1=3.85\times 10^6$ s⁻¹, $\gamma_2=1.5\times 10^{-12}$ cm³ s⁻¹, and $\gamma_3=7\times 10^{-31}$ cm⁶ s⁻¹. Filled squares are the minority-carrier lifetimes in differently doped n -type 4H-SiC, the broken line is $\gamma_3^{eh}=5\times 10^{-31}$ cm⁶ s⁻¹ quadratic fit.

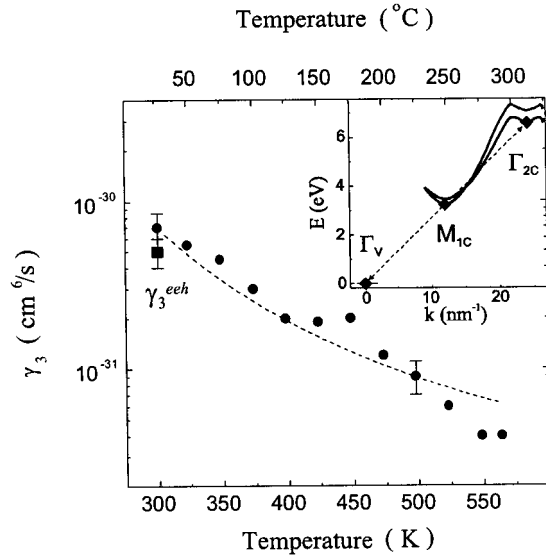


FIG. 4. Temperature dependence of Auger recombination coefficient in 4H-SiC: circles are measured data, dashed line is simulated considering variable Auger threshold energy. The single square indicates room temperature value of the e - e - h AR coefficient measured in highly doped n -type 4H-SiC. The inset shows a diagram of direct e - e - h Auger transitions along the Γ - M conduction band segment of the Brillouin zone.

+ γ_3^{hhe} . The estimated value of $\gamma_3 = 7 \times 10^{-31} \text{ cm}^6 \text{ s}^{-1}$ at 300 K is comparable to what is observed in other indirect band-gap semiconductors under high-excitation conditions, e.g., $3.8 \times 10^{-31} \text{ cm}^6 \text{ s}^{-1}$ in Si,³ and $1.1 \times 10^{-31} \text{ cm}^6 \text{ s}^{-1}$ in Ge.⁴ The AR coefficients at elevated temperatures were extracted by fitting simulated and experimental $\tau(N)$ dependencies in a similar way. The main results of the AR variation with temperature are summarized in Fig. 4. A pronounced reduction of the AR coefficient from $(7 \pm 1) \times 10^{-31} \text{ cm}^6 \text{ s}^{-1}$ to $(4 \pm 0.5) \times 10^{-32} \text{ cm}^6 \text{ s}^{-1}$ within a temperature interval 300–565 K is observed.

Finally, we have estimated the AR rate in a series of heavily doped n -type 4H-SiC samples. The e - e - h type Auger transition is considered to be the predominant carrier recombination mechanism for high donor concentrations N_D , thus AR coefficient can directly be derived from the relationship $\gamma_3^{eeh} = [\tau N_D^2]^{-1}$. This method, however, provides only an upper limit for the γ_3^{eeh} value, since high concentrations of impurities could, in principle, both distort the band structure and introduce additional defect related recombination channels. The low-injection lifetimes of differently doped samples are displayed in Fig. 3 by solid squares. The minority-carrier lifetime decreases quadratically with doping concentration and implies $\gamma_3^{eeh} = (5 \pm 1) \times 10^{-31} \text{ cm}^6 \text{ s}^{-1}$ at room temperature. It should be noted that a prominent impurity-assisted Auger process, which in heavily doped material would not be surprising and could involve the shallow donor centers, must be considered more carefully. Further complementary studies of high p -type doped 4H-SiC material could possibly clear up this contribution.

Preliminary inspection of the 4H-SiC band structure^{10,11} confirms favorable conditions for the e - e - h AR process in contrast to the h - h - e type in the valence band. A schematic of possible phonon-less e - e - h Auger transitions along the Γ - M conduction band segment on the Brillouin zone is

shown in the inset to Fig. 4. The high-energetic electron is excited by the umklapp process from the M_{1c} to the lowest Γ_{2c} point of the same zone, obeying both energy and momentum conservation requirements. In parabolic band approximation, the threshold of the e - e - h transition can be expressed analytically⁷ as $E_{th} = A(2E_g^{ind} - E_g^{dir})$, where $A = m_{d\parallel} / (m_{v\parallel} + 2m_{c\parallel} - m_{d\parallel})$. Assuming the values of longitudinal effective masses $m_{d\parallel} \approx 0.3m_0$ at the Γ_{2c} point,⁸ $m_{c\parallel} = 0.58m_0$ at the M_{1c} point,⁹ and $m_{v\parallel} \approx 4.2m_0$ in the valence band,⁵ the direct band-gap $E_g^{dir} = 6.53 \text{ eV}$,⁵ and the indirect band-gap $E_g^{ind} = 3.275 \text{ eV}$,⁶ the e - e - h AR threshold can be roughly evaluated as $E_{th} = 1.2 \text{ meV}$. In view of possible uncertainties in the band-gap values, the analyzed transition can also be considered as a zero-threshold process at room temperature. The established $\gamma_3(T)$ dependence is opposing what could be expected. In fact, theories of phonon-less AR with $E_{th} = \text{const.}$ or phonon-assisted AR mechanism, both describe a general increase of the recombination rate with temperature.¹⁰ A likely explanation of our experimental results is an increase of the AR threshold with temperature. A simplified assumption of the linear $E_{th}(T)$ variation in the expression for direct band-to-band AR coefficient $\gamma_3(T) = \gamma_{30} \exp[-E_{th}(T)/k_B T]$ qualitatively explains the reduction of γ_3 at high temperatures (see Fig. 4). The presented $\gamma_3(T)$ fit is simulated considering $E_{th}(T) = (0.45 \text{ meV/K}) \times (T - 300 \text{ K})$ and $\gamma_{30} = 7 \times 10^{-31} \text{ cm}^6 \text{ s}^{-1}$. The different narrowing of direct and indirect band gap with temperature is presumably responsible for the observed quenching of e - e - h AR process.

In summary, applying an optical pump-probe technique we have observed and quantified Auger recombination in 4H-SiC epilayers and in heavily doped substrates. Our calculation also indicates a sizable bimolecular recombination coefficient of $\gamma_2 = 1.5 \times 10^{-12} \text{ cm}^3 \text{ s}^{-1}$. Ultimately, the observed reduction of Auger recombination rate with temperature implies an important practical conclusion that higher temperatures are favorable for operation of 4H-SiC based power devices.

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